

what the bits mean

digitalisierung und programmierung · prof. dr. nicolas meseth

- 1 the agreement is everything
- 2 build your own codebook
- 3 where does a character end?
- 4 the same bits, other meanings
- 5 codes without computers

part 1

the agreement is everything

who still knows
what this
means?

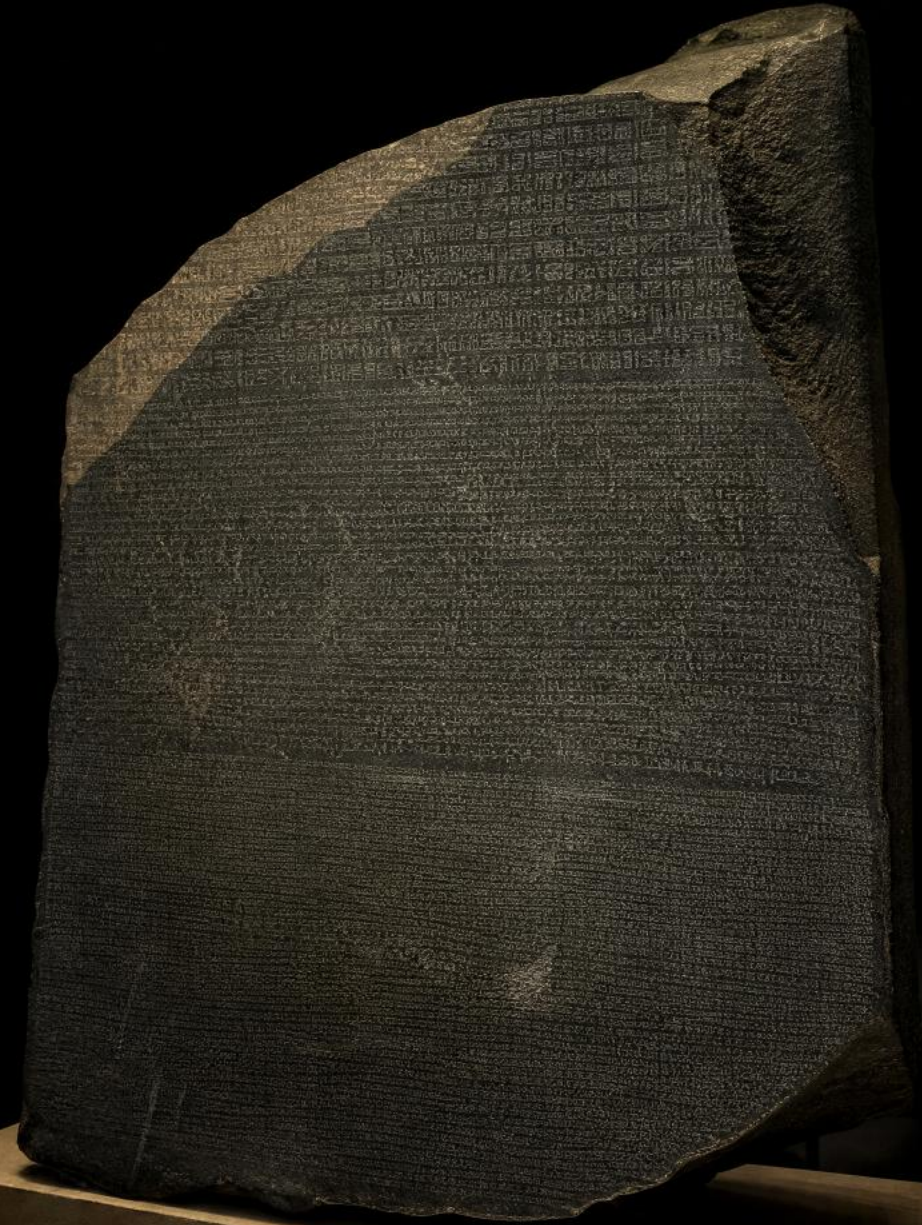


image: ai-generated (gpt-image-2)

the agreement is everything

what a code system is

character		number	bits
A	→	65	
B	→	66	
C	→	67	
a	→	97	

a code is not encryption, not compression, and not source code

a letter is a number. the letter comes back only when someone draws it again.

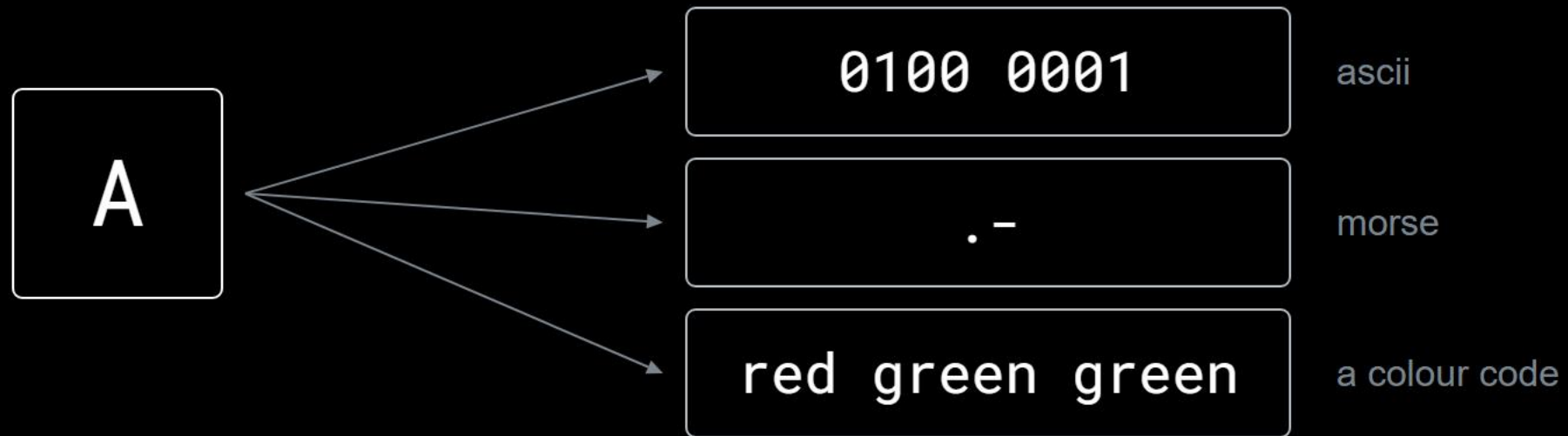
what a code system is

character		number	bits
A	→	65	0100 0001
B	→	66	0100 0010
C	→	67	0100 0011
a	→	97	0110 0001

a code is not encryption, not compression, and not source code

a letter is a number. the letter comes back only when someone draws it again.

the same letter, three code systems



agreed, not given by nature. that is why there are several.

when the table is too small

does ascii have it?

A	z	7	?	ä	€	π	
✓	✓	✓	✓	✗	✗	✗	

utf-8 then writes those numbers down: common ones in one byte, rare ones in up to four

one table was too small. so a bigger one was agreed.

when the table is too small

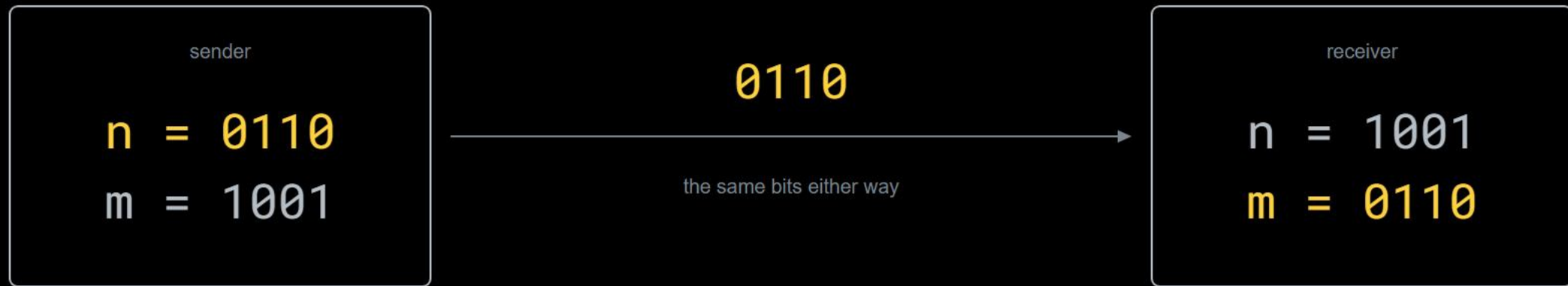
does ascii have it?

A	z	7	?	ä	€	π	
✓	✓	✓	✓	✗	✗	✗	
65	122	55	63	228	8364	960	

utf-8 then writes those numbers down: common ones in one byte, rare ones in up to four

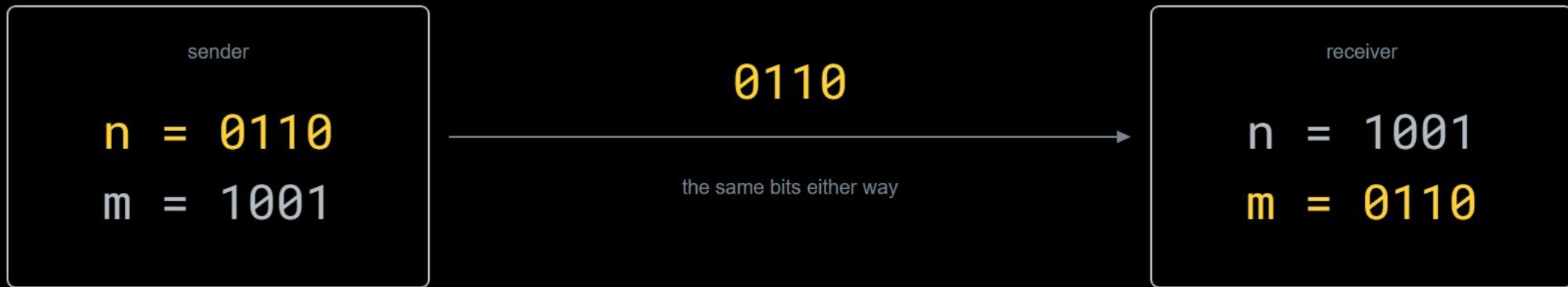
one table was too small. so a bigger one was agreed.

two tables, one message



no error message. just clean nonsense.

two tables, one message



typed

sonne

no error message. just clean nonsense.

two tables, one message



no error message. just clean nonsense.

part 2

build your own codebook

how many positions per letter?

your alphabet



26 letters to encode

1 position

4 codes

×

not enough for 26

2 positions

$4 \times 4 = 16$

3 positions

$4 \times 4 \times 4 = 64$

26 letters, 3 colour positions, 38 codes to spare.

how many positions per letter?

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enough for 26

26 letters, 3 colour positions, 38 codes to spare.

the codebook

a 000	b 001	c 002	d 003	e 010	f 011	g 012	h 013	
i 020	j 021	k 022	l 023	m 030	n 031	o 032	p 033	
q 100	r 101	s 102	t 103	u 110	v 111	w 112	x 113	
y 120	z 121	 122	123	0 130	1 131	2 132	3 133	
4 200	5 201	6 202	7 203	8 210	9 211	212	213	
220	221	222	223	230	231	232	233	
300	301	302	303	310	311	312	313	
320	321	322	323	330	331	332	333	

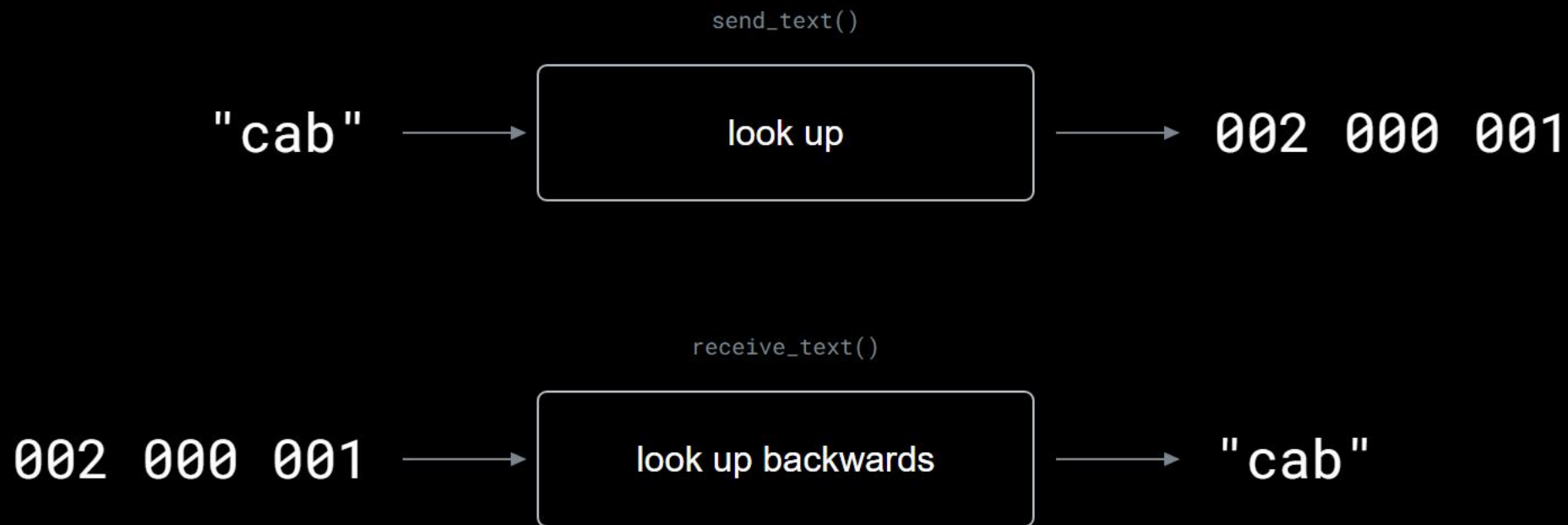
the spare codes are stock, not waste.

the same table, both directions



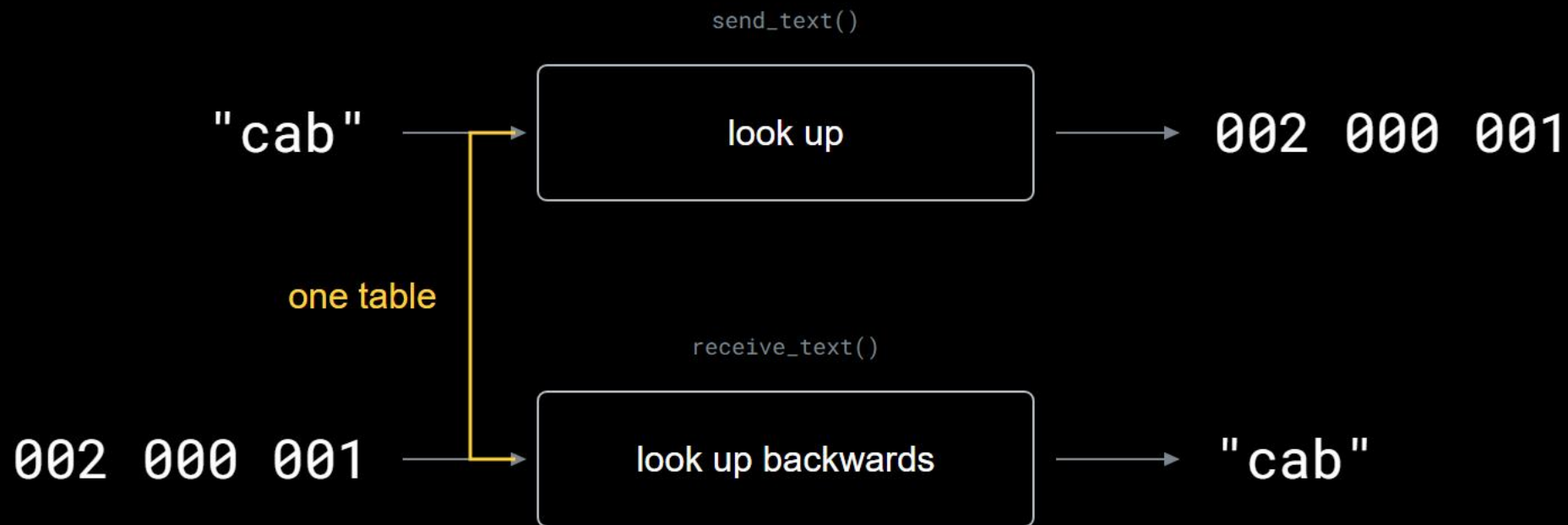
decoding is the same table, read the other way.

the same table, both directions



decoding is the same table, read the other way.

the same table, both directions



decoding is the same table, read the other way.

what is not in the table

```
send_text("guten morgen")
```

your codebook has a to z and nothing else

g	u	t	e	n	space	m	o	r	g	e	n
✓	✓	✓	✓	✓	?	✓	✓	✓	✓	✓	✓

three ways out, and you have to pick one

add it
to the table

drop it
before sending

stop
with a message

something happens either way. decide which.

part 3

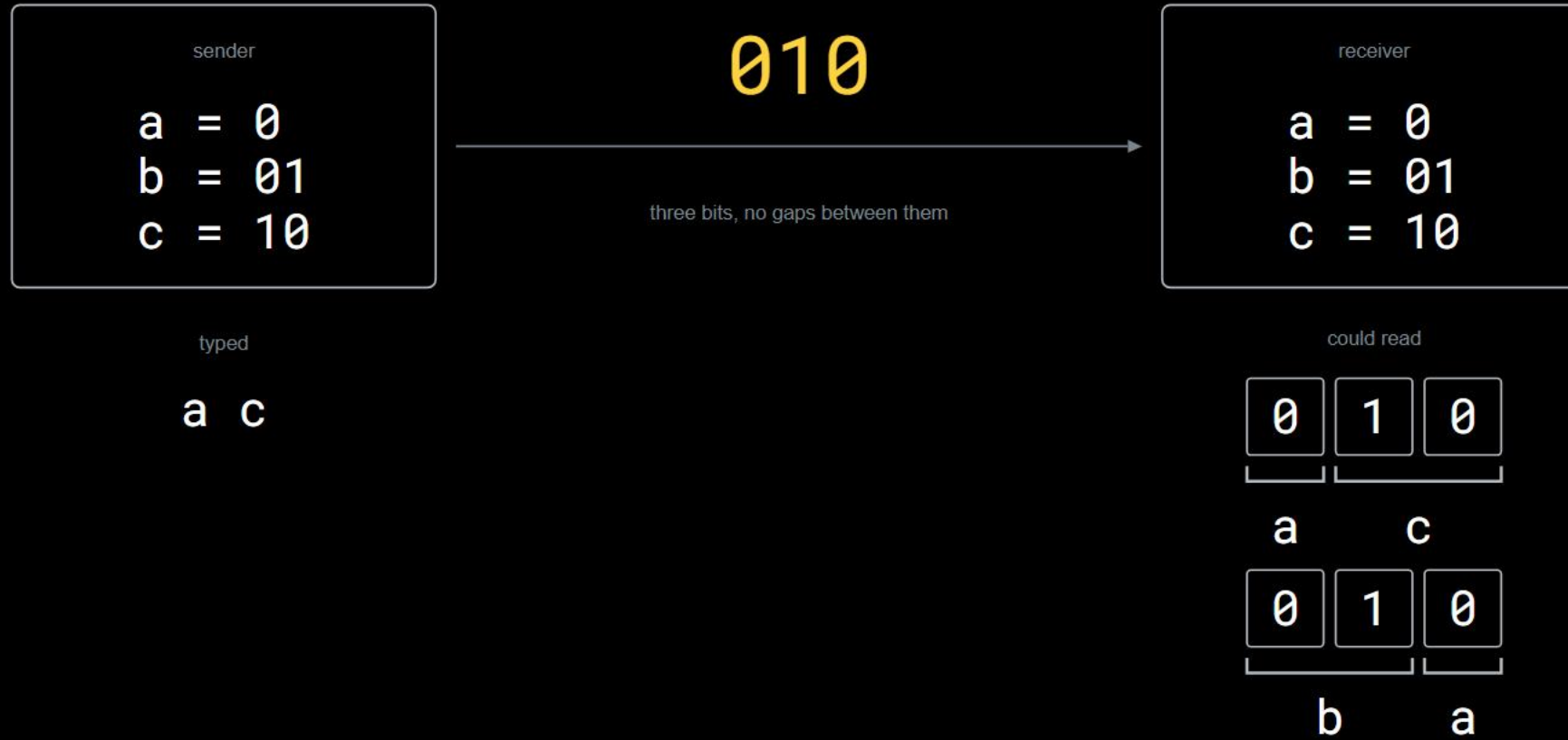
where does a character end?

a code that does not work



same bits, two readings. the receiver cannot see the boundary.

a code that does not work



same bits, two readings. the receiver cannot see the boundary.

three ways out

fixed length

a = 00

b = 01

c = 10



count instead of guessing

a separator

a = .-

b = -...

c = -.-.



morse pauses between letters

no code word
starts another

a = 0

b = 10

c = 110



read until it matches

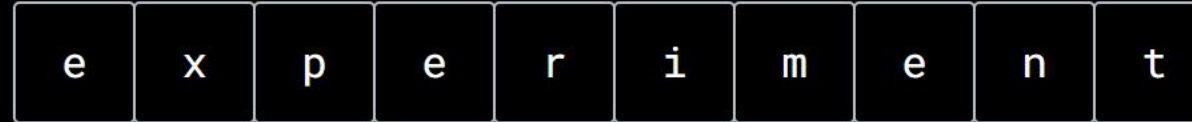
three answers to one question. all of them work.

what fixed length costs

the same word in two codes of your own

fixed

three positions for every letter



30 positions

3.0 per letter

variable

two for the common ones, four for the rest



26 positions

2.6 per letter

the ten most common letters in german and english: e n i s r a t d h u

shorter on average, not in every case.

where does a character end?

part 4

the same bits, other meanings

one byte sequence, three readings



the photo

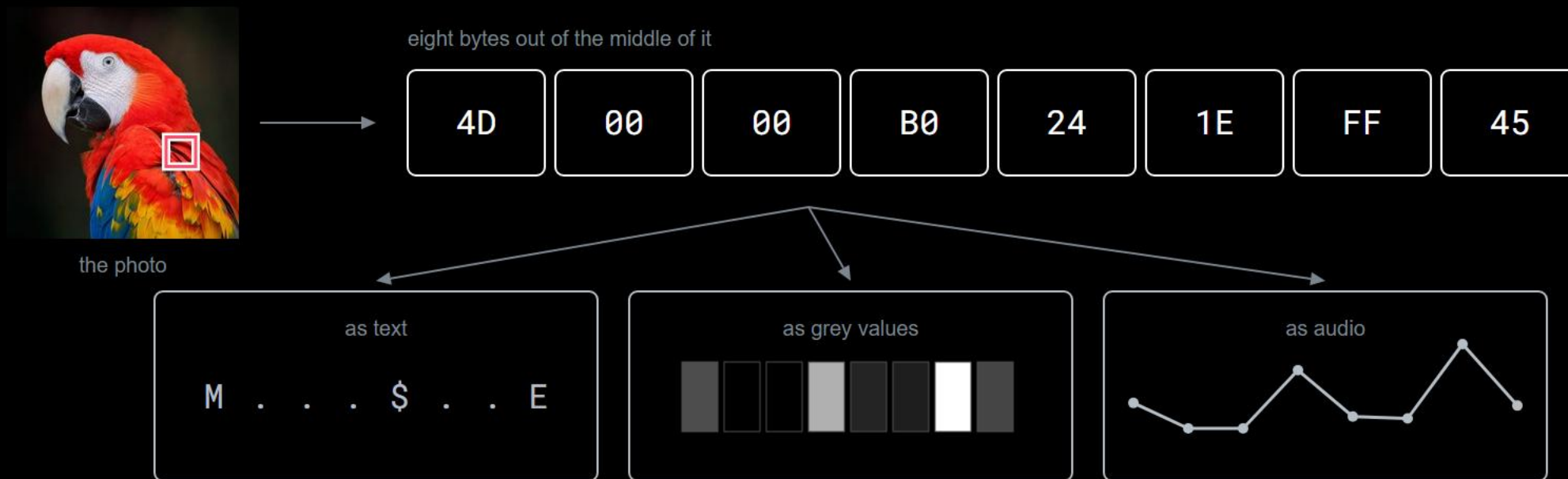
eight bytes out of the middle of it



the bytes do not say which reading is right.

the same bits, other meanings

one byte sequence, three readings



the bytes do not say which reading is right.

one pixel, three numbers



one pixel of it



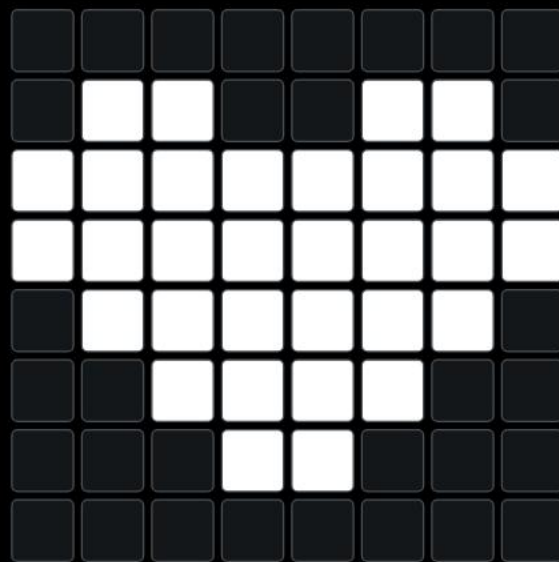
the same colour, written as hex: two places per byte

#ECAB03

the same three numbers your sender writes: `led.set_color(236, 171, 3)`

the same bits, other meanings

try it: pixel painter

	binary	hex
	0000 0000	00
	0110 0110	66
	1111 1111	FF
	1111 1111	FF
	0111 1110	7E
	0011 1100	3C
	0001 1000	18
	0000 0000	00

64 pixels = 64 bits = 8 bytes · 32 pixels on

start from:



break a byte

flip one bit

repair

invert

clear

the same bits, other meanings

who says which reading?

the file extension

.txt .png .wav

a header in the file

the first bytes say it

a type field in your frame

you decide what it says

challenge 3

the type is another agreement.

part 5

codes without computers

four agreements, no computer



barcode

bar widths → digits



sheet music

position → pitch and length



braille

six dots → a character



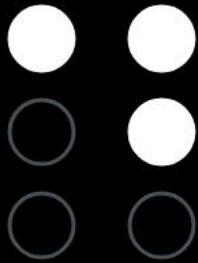
dna

three bases → an amino acid

none of them needs a computer. all of them need an agreement.

the same calculation, three times

braille



2 states, 6 positions

$$2^6 = 64$$

dna



4 bases, 3 positions

$$4^3 = 64$$

your alphabet



4 colours, 3 positions

$$4^3 = 64$$

braille, dna and your alphabet all landed on 64.

write it down.
twice.

